दक्षिण बिहार केन्द्रीय विश्वविधालय

Central University of South Bihar

Question Paper of End-Term Examination-Dec. 2025

Name of Department: Computer Science
Programme: M.Sc(AI)
Session: 2025-27
Semester: I
Course Title: Mathematical Foundation of AI
Course Code: CAI81DC00304
Course Credit: 4
Duration: 3 Hours
Enrollment Number:
Max Marks: 70

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis. Use of a calculator is not allowed.

Section - A

Answer all the questions in not more than 50 words. Each question carries 1 Points. $(1 \times 10 = 10$ Points)

1. Write De Morgan's law.
2. Write deduction theorem.
3. Define a positive definite matrix.
4. Write $ax^2 + 2bxy + cy^2$ where x, y are variables in terms of the multiplication of a constant matrix and two vectors of the variables.
5. Find the number of multiplication and addition we do when we multiply two matrices, each of size $n \times n$.
6. Write the elimination matrix of size 3×3 which does the row transformation $R_2 \leftarrow R_2 - 2R_1$.
7. Let $p(x, y) = P(X = x, Y = y)$ be the joint probability distribution function for the discrete random variables X and Y. Define the probability distribution function for the discrete random variable X.
8. State Chebyshev's inequality when the sample mean and the sample standard deviation are given for a data set.
9. State all the axioms of probability.
10. If the events A and B are independent then $P(A|B) =$?

Section - B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points. $(6 \times 5 = 30$ Points)

1. Compute the LU decomposition for the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}$.
2. By using deduction theorem or any other way, show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S), \neg R \vee P,$ and Q.
3. Explain quantifiers, free variable, and bound variable with an example for each.
4. Find the null space of the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 2 & 6 & 9 \\ -1 & -3 & 3 \end{bmatrix}$.

5. Define the Bernoulli, binomial, and normal random variables, and also compute the mean and variance of Bernoulli and binomial random variables.

6. In answering a question on a multiple-choice test, a student either knows the answer or he/she guesses. Let p be the probability that he/she knows the answer and $1-p$ the probability that he/she guesses. Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple choice alternatives. What is the conditional probability that a student knew the answer to a question, given that he/she answered it correctly?

Section - C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points.

$(2 \times 15 = 30$ Points)

1. Find the orthogonal vectors from $a^T = (1,1,2)$, $b^T = (1,-1,0)$, and $c^T = (1,0,4)$ using Gram-Schmidt orthogonalization.

2. Compute the inverse of the following matrix using Gauss-Jordan elimination $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

3. The density function of X is given by

$$f(x) = \begin{cases} a + bx^2 & \text{if } 0 \leq x \leq 1 \\ 0 & Otherwise \end{cases}$$

If $E[X] = \frac{3}{5}$ then find a and b. Then compute Var(X).